

STRATEGIC PRODUCT VULNERABILITY? CHINA LEAST, THEN EUROPE AND THEN THE US

Koen De Leus

“Germany is too dependent on the US for its security, on Russia for its energy and on China for its exports.” That was, in essence, Brookings’ Constanze Stelzenmüller’s diagnosis in June 2022, and it was – and is – valid as well for Europe as a whole.

But does this dependency also apply to technological and industrial products? The European Commission’s EXternal Vulnerability Index (EXVI) answers that very question, mapping out the EU’s exposure to foreign supply chains.

EXVI is built on two pillars, with two simple questions for each product: 1/ How dependent is a region on a handful of foreign suppliers? 2/ How globally competitive is it in that product measured by its imports versus exports? The result is a single score from 0 (low vulnerability) to 1 (high). A region scores high (vulnerable) when there is both a heavy reliance on others and a weak market position.

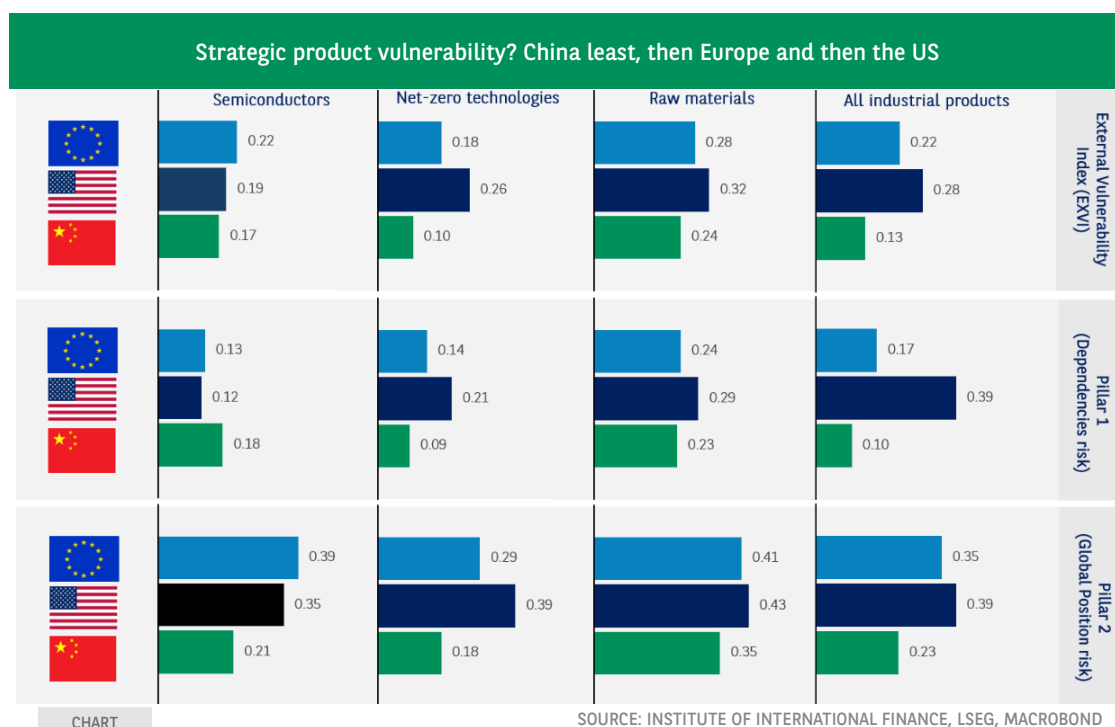
Relatively good results for the EU

The EU is relatively well positioned, with a score of 0.18 on net-zero technologies and 0.22 on semiconductors. Raw materials appear to be the weakest point (0.28). Interestingly, compared to its peers, the EU is more vulnerable than China across all three chains (0.13), but significantly less vulnerable than the US overall (0.22 vs. 0.28). The only sector where the EU score is weaker than the US one is on semi-conductors, and the difference is not large (0.22 vs 0.19). Like the EU, the US’ vulnerability is biggest in raw material (0.32). This is, among other factors, due to outsourcing polluting mining starting in the 1990s. Looking at the essential components for the green transition, the EU score on raw materials is concerning, but the better one on net-zero technologies is more encouraging.

The EU challenge is not its dependencies but its relatively weak global position

Taking a closer look at the two pillars across the different products, it becomes clear that Europe’s problem is mainly related to its global position (Pillar 2). Dependency (Pillar 1) itself is not the core issue, with much lower scores than the ones for Pillar 2. The EU dependency risk on raw materials is even comparable to China score (0.24 vs 0.23), and the EU score on semi-conductors (0.13) is the lowest of the other results for the region.

On the other hand, Pillar 2 scores are more negative, illustrating the lack of EU competitiveness in these areas, with scores hovering between 0.29 for net-zero technologies up to a high 0.39 for semiconductors and 0.41 for raw materials. Europe simply does not produce enough of these products at competitive prices, as big European companies in these sectors are limited. Take semiconductors, for example. Imports are not extremely



concentrated, as shown by the relatively low Pillar 1 score of 0.13. However, the high Pillar 2 score of 0.39 shows that the EU is weak at producing semiconductors itself. That is precisely why the European Chips Act is focusing on building domestic production capacity rather than on diversifying imports. The same reasoning applies to raw materials and, to a lesser extent, to net-zero technologies.

One trend is undeniable: China consistently enjoys the lowest vulnerability. Decades of strategic plans, as well as subsidies and other practices to restrict open markets, have enabled it to build up domestic supplies and stocks.

The EU's multiple responses

Looking ahead, the EU can deploy a number of options in response, from circularity to domestic mining, including international partnerships. Brussels is aiming to mine or recycle 10% of critical materials by 2030, but there is paradox at play here, as circularity only works once sufficient volumes have first been mined and used. Therefore, in the coming years, international partnerships will be a priority and, for that matter, the EU has been very active on that front since the start of the second Trump presidency.

The EU is aware of the challenges that it faces. And the EXVI indices are particularly relevant for assessing supply chains targeted by major policy initiatives, such as the Chips Act (strengthening the EU's semiconductor ecosystem and technological sovereignty), the Net Zero Industry Act (NZIA) (building domestic manufacturing capacity for clean-tech products such as solar, wind and batteries) and the Critical Raw Materials Act (CRMA) (securing and diversifying the EU's supply of critical and strategic raw materials). The challenges have been identified and an action plan has been drawn up. Now it needs to be implemented on the ground.

1 The latest full report is dated January 2025 and can be found here: [EXternal Vulnerability Index \(EXVI\) - Internal Market, Industry, Entrepreneurship and SMEs](#).

koen.deleus@bnpparibasfortis.com



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